

Test case management system for robotic

Functional requirement

- able to add test case
- able to modify the test case
- able to view single test case
- able to view all the test case
- able to delete test case
- able to start the test run,
- able to check test run result
- able to view all the test run
- able to check HW status

Non-functional requirement

- need to handle at least 100 people able to view all test case and view test result
- test case is less than 500 data in DB
- test run could have 200k data (assuming 100 people submit 5 test run a day, data save up to 1 year)
- when start new test run, need to check HW available and return msg if HW is in use or not available
- test case display latency should be <500ms
- Longest test run within 10min, if user check result but test run is not finish, should mark status as "running"

API design

1. Add test case

Post /new-test-case

```
{
  Name: ""
  Description: ""
  HWConfigPath: ""
  ExecuteSteps: ""
  ExpectedResult
} ->
{
  testID
  Name: ""
  Description: ""
  ...
}
```

2. Update test case

PUT /update-test-case/{testID}

```
{
  Name: ""
  Description: ""
  HWConfigPath: ""
  ExecuteSteps: ""
}
```

ExpectedResult

} ->

{

testID

Name: ""

Description: ""

...

}

3. Get test case list

GET /list-test-case

->

{

testID: 1

...

testID: 2

...

}

4. View single test case

GET /list-test-case/{testID} ->

{

testID

Name: ""

Description: ""

...

}

5. delete test case

DELETE /delete-test-case/{testID}

6. Start test case

POST /start-test/{testID}

->

{

testID:

unique-executID:

Name:

...

status: RUNNING | HW in use | HW disconnected | COMPLETE | ERROR

msg:

}

7. Access test case result

GET /view-test-result/{unique-executID}

->

{

testID:

unique-executID:

Name:

```
...
status: RUNNING | HW in use | HW disconnected | COMPLETE | ERROR
msg:
}
```

8. Get test run list

GET /list-test-run

->

```
{
  unique-executelD:: 1
  ...
  unique-executelD:: 2
  ...
}
```

9. Get HW status

GET /hardware-status ->

```
{
  <hw01>: {
    name
    model:
    connection: true | false
    in_use: true | false
  }
  <hw02>:
  ...
}
```

10. HW config:

Needs to be a yaml file save in DB, each HW provide a yaml config, and save it in a path where api/server can access. Put some example below:

motor.yaml:

IP: ...

port: ...

speed: ...

...

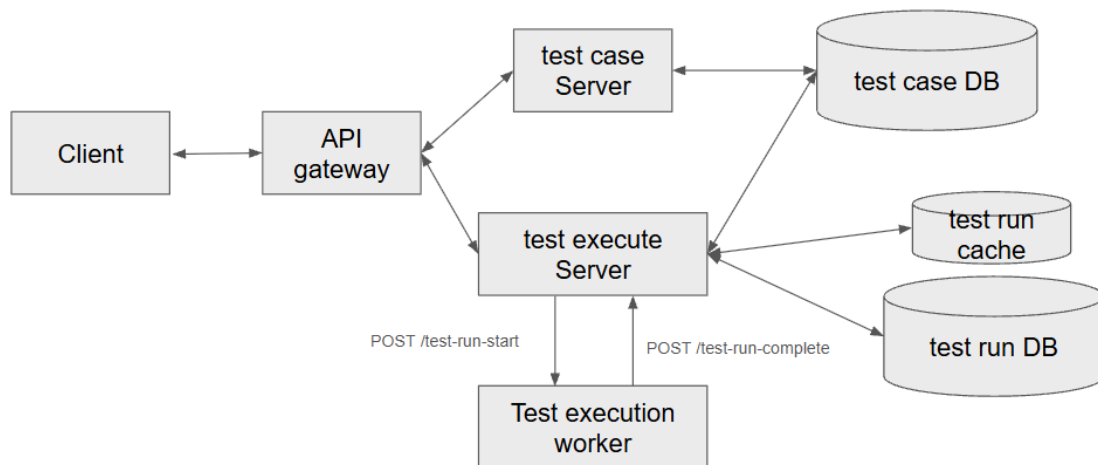
camera.yaml

IP: ...

port: ...

filter: ...

High level design



1. Test case server should be able to handle 100 + reading test case DB. Assume test case is less than 500 data
2. Test run data can be up to 200k, add test run cache if people want to frequently read test result
3. Test run cache needs to set different TLT based on status, if "running" use 10 sec, if "complete" or "error", use 1 hour